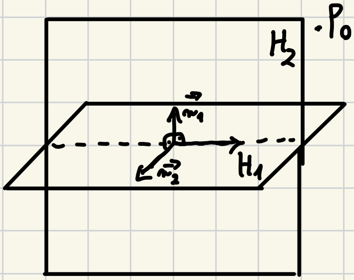
$P_0(1,-1,1)$, $H_1: x-y+z=0$ $H_2: 2x+y+z+1=0$

$\vec{n}_1=[1,-1,1]\perp H_1$ $\vec{n}_2=[2,1,1]\perp H_2$



$n_1 \times n_2 = [1,-1,1]\times[2,1,1] =$

$= [-2,1,3] \perp$ do szukanej płaszczyzny

$H: -2(x-1)+1(y+1)+3(z-1)=0$

$H: -2x+y+3z=0$

$l: \frac{x}{2} = 1-y = \frac{z+1}{2}$

$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$

$\frac{x-0}{2} = \frac{y-1}{-1} = \frac{z-(-1)}{2}$

$P(0,1,-1)\in l$

$\vec{k}=[2,-1,2] \parallel l$

$H: x+y+z-1=0$

$l \begin{cases} x=2t \\ y=1-t \\ z=-1+2t, \ t\in\mathbb{R} \end{cases}$

$2t+1-t-1+2t-1=0$

$3t-1=0$ $t=\frac{1}{3}$

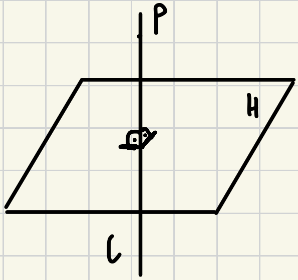
$P_1\left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)$

$P = H \cap l$

Zad. 2

Napisać równanie prostej o współrzędnej $P(1,-2,3)$ na płaszczyźnie:

$H: 3x-4y+2z-6=0$



$\vec{n} = [3,-4,2] \perp H$

$(l \perp H) \wedge (\vec{n} \perp H) \Rightarrow \vec{n} \parallel l$

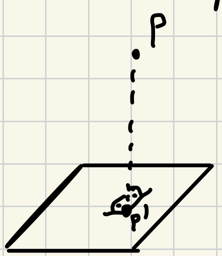
$$l \begin{cases} x = 1+3t \\ y = -2-4t \\ z = 3+2t, \; t \in R \end{cases}$$

$$l: \frac{x-1}{3} = \frac{y+2}{-4} = \frac{z-3}{2}$$

Zad. 3

rzut prostokątny

Znaleźć rzut punktu $P(2,1,1)$ na płaszczyznę $H: x+y+3z+5=0$



Wyznaczamy prostą l taką, że:

$l \ni P \wedge l \perp H$

$\vec{n} = [1,1,3] \perp H$

$$l \begin{cases} x = 2+t \\ y = 1+t \\ z = 1+3t, \; t \in R \end{cases}$$

$P' = l \cap H$

$2+t+1+t+3(1+3t)+5=0$

$2t+3+3+9t+5=0$

$11t=-11$

$t=-1$

$P'(1,0,-2)$

Wyznaczyć punkt P_1, który jest symetryczny do punktu P względem płaszczyzny H

$P(2,1,1) \quad P'(1,0,-2)$

P'

$P \quad |PP'| = |P'P_1| \quad P_1$

$$\frac{x_P + x_{P_1}}{2} = x_{P'} \qquad x_{P_1} = 2x_{P'} - x_P = 2-2=0$$

$$\frac{y_P + y_{P_1}}{2} = y_{P'} \qquad y_{P_1} = 2y_{P'} - y_P = -1$$

$$\frac{z_P + z_{P_1}}{2} = z_{P'} \qquad z_{P_1} = 2z_{P'} - z_P = -5$$

$$\boxed{P_1(0,-1,-5)}$$

Zad. 4

Znaleźć równanie płaszczyzny, która zawiera $P(2,-3,1)$ i

$l: \frac{x-1}{5} = y+3 = \frac{z}{2}$

$l: \begin{cases} \frac{x-1}{5} = y+3 \\ y+3 = \frac{z}{2} \end{cases}$ $l: \begin{cases} x-1 = 5y+15 \\ 2y+6 = z \end{cases}$

$l: \begin{cases} x-5y-16=0 \\ 2y-z+6=0 \end{cases}$

$H_\lambda: x-5y-16+\lambda(2y-z+6)=0$

Wstawiamy punkt P

$2+15-16+\lambda(-6-1+6)=0$

$1-\lambda=0$

$\lambda=1$

$H: x-5y-16+2y-z+6=0$

$H: x-3y-z-10=0$

Zad. 5

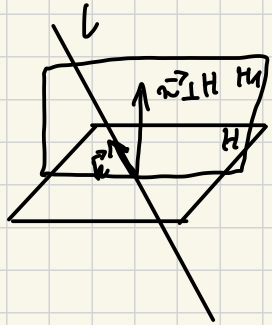
Napisać równanie płaszczyzny zawierającej prostą

$l: \frac{x-2}{2} = y-3 = \frac{z+1}{2}$ i prostopadłej do płaszczyzny

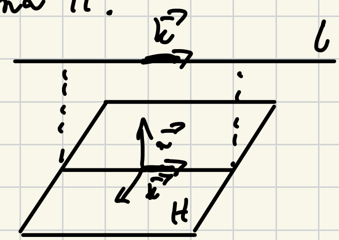
$H: x+2y-3z+7=0$ oraz wyznaczyć rzut prostej

l na H.

I

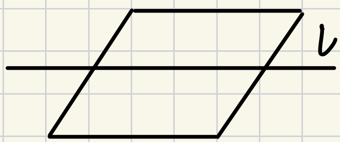


II

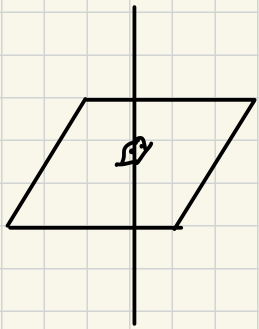


Przypadki I i II na kolosie

III



IV



$P(2,3,-1) \in l \subset H_1$

$\vec{k} = [2,1,2] \parallel l$

$\vec{n} = [1,2,-3] \perp H$

$$\vec{k} \times \vec{n} = \begin{array}{r} [2,1,2] \\ \times [1,2,-3] \\ \hline [-7,8,3] \end{array} \perp H_1$$

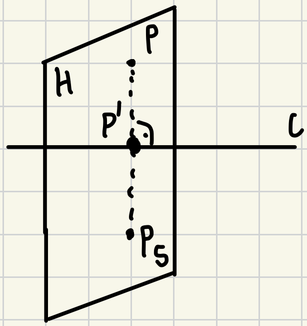
$H_1: -7(x-2)+8(y-3)+3(z+1)=0$

$H_1: -7x+8y+3z-7=0$

$$l': \begin{cases} -7x+8y+3z-7=0 \\ x+2y-3z+7=0 \end{cases}$$

Zad. 6

P_s $\quad P(4,3,10)$ $\quad l: \frac{x-1}{2} = \frac{y-2}{4} = \frac{z-3}{5}$



$H: (P \in H) \wedge (H \perp l)$

$(l \perp H) \wedge (\vec{k} \parallel l) \Rightarrow \vec{k} \perp H$

$\vec{k} = [2, 4, 5]$

$$H: 2(x-4) + 4(y-3) + 5(z-10) = 0$$

$$H: 2x + 4y + 5z - 70 = 0$$

P' – rzut pkt. P na l

$P' = H \cap l$

$$l: \begin{cases} x = 1 + 2t \\ y = 2 + 4t \\ z = 3 + 5t, \ t \in R \end{cases}$$

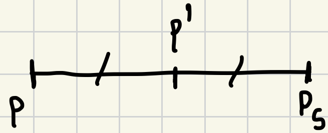
$$2(1+2t) + 4(2+4t) + 5(3+5t) - 70 = 0$$

$$2 + 4t + 8 + 16t + 15 + 25t - 70 = 0$$

$$45 = 45t$$

$$t = 1$$

$P'(3,6,8)$



$\frac{4+x}{2} = 3 \quad \frac{3+y}{2} = 6 \quad \frac{10+z}{2} = 8$

$x = 2 \quad y = 9 \quad z = 6 \quad P_s(2,9,6)$

$$l_1: \frac{x-9}{4} = \frac{y+2}{-3} = z$$

$$l_2: \frac{x}{-2} = \frac{y+7}{9} = \frac{z-2}{2}$$

$P_1(9,-2,0) \in l_1$

$\vec{k_1} = [4,-3,1] \parallel l_1$

$P_2(0,-7,2) \in l_2$

$\vec{k_2} = [-2,9,2] \parallel l_2$

GEOGEBRA

MAXIMA

OCTAVE

WOLFRAMALFA

$$\vec{P_1P_2} = [-9,-5,2]$$

$$\det P = \det \begin{bmatrix} \vec{P_1P_2} \\ \vec{k_1} \\ \vec{k_2} \end{bmatrix} = \begin{array}{c} \begin{vmatrix} -9 & -5 & 2 \\ 4 & -3 & 1 \\ -2 & 9 & 2 \end{vmatrix} \end{array} \begin{array}{l} w_1 - 2w_2 \\ = \\ w_3 - 2w_2 \end{array}$$

$$= \begin{vmatrix} -17 & 1 & 0 \\ 4 & -3 & 1 \\ -10 & 15 & 0 \end{vmatrix} = 1 \cdot (-1)^{2+3} \begin{vmatrix} -17 & 1 \\ -10 & 15 \end{vmatrix} =$$

$$= -(-17 \cdot 15 + 10) = 245 \neq 0 \Rightarrow \text{proste skośne}$$